

How Much Real PHY Does a 5G Digital Twin Need?

Two Constants, Measured on OpenAirInterface

Abstract—Network digital twins are where O-RAN rApps are validated before they touch production, so the twin’s link-to-system (L2S) abstraction—the map from SINR to throughput—decides which controllers look good. Received wisdom says this map must be measured on a real physical layer. We ask *how much* of a real PHY a twin needs, and answer by measuring two constants on OpenAirInterface (OAI). Our open, CPU-only, multi-cell twin (OpenStreetMap $\rightarrow$ Sionna RT $\rightarrow$ per-RE SINR $\rightarrow$ EESM $\rightarrow$ OAI’s `nr_dlsim` curve) hosts a proportional-fair traffic-steering rApp and three bridges to a live OAI 5G Standalone stack, including per-cell ray-traced channel injection on a two-DU stack whose waveforms are physically summed. Over three cities $\times$ eight layouts the rApp lifts median per-user throughput by $12.0 \pm 12.8\%$ and served cell-edge throughput by $28.6 \pm 37.2\%$. We separate a twin’s *reporting* error from its *decision* error, the latter measured as rApp *regret*: the utility lost by planning on a surrogate twin and living in the OAI-grounded one. An uncalibrated Shannon curve inflates throughput 41% and costs 0.76 nats; a single fitted implementation margin $\delta=4.71$ dB reduces these to 1.5% and 0.04 nats, and the curve’s shape buys nothing further. The customary “decisions differ” statistic (33–42% for every surrogate) cannot separate them; only regret can. On the live stack the twin’s link rate exceeds what OAI’s MAC schedules by a *constant* factor ($\kappa=0.470 \pm 0.004$, CV 0.8% over MCS 0–28), so it cancels in ratios: the twin’s *relative* per-UE steering-gain prediction matches the real MAC to 1.2 ± 5.7 points. Delivered goodput lies a further 80% down and grows with SINR—the emulator’s ceiling, not the twin’s error. Finally, a null control on two *identical* real cells shows the natural goodput probe reporting $-21.0 \pm 1.4\%$ where truth is zero; we use it to retract a multi-cell gap we previously reported.

Index Terms—Digital twin, 5G NR, OpenAirInterface, link-to-system abstraction, O-RAN, rApp, traffic steering.

I. INTRODUCTION

The digital twin is becoming the environment in which AI/ML-driven RAN control—O-RAN rApps and xApps—is trained and validated before it touches production [1]. Its value hinges on data fidelity: a traffic-steering rApp decides which cell serves each user from signal, interference and load, which depend jointly on 3D propagation (site-specific), on neighbouring cells (multi-cell), and on a link model turning SINR into a real modulation-and-coding outcome (protocol-stack realism). Evaluating such a controller needs all of these, across many layouts rather than one scenario: our own single-seed pilot suggested a uniformly large rApp gain, which a 24-run study tempered into a robust-direction, variable-magnitude result.

Open tooling splits into camps that do not meet. System-level simulators scale to many cells but abstract the physical (PHY) layer with idealized SINR-to-throughput curves. OAI-based emulators execute the real 5G-NR stack but, with realistic propagation, cover a single link. Commercial twins close the rApp loop but are proprietary. No open platform combines a real protocol stack, a site-specific ray-traced channel, a multi-cell network, and a closed control loop (Table I).

A subtler gap concerns the *L2S abstraction* itself. System-level evaluation cannot afford full PHY processing per user, so it relies on a mapping from SINR to a modulation-and-coding outcome. Open simulators use idealized or 3GPP-calibrated curves; we obtain ours from OAI’s real receiver (`nr_dlsim` with LDPC decoding and rate matching). At 10 dB SINR the OAI-measured curve yields 2.16 bits/s/Hz against Shannon’s 3.46, a 38% implementation loss, and its MCS steps are non-uniform, spanning up to 2.2 dB.

The obvious claim is that this curve is therefore necessary. We tried to make it, and the evidence did not support it. Measuring a real PHY is expensive, so the useful question is not *whether* an idealized curve is wrong—it plainly is—but *which part* of the real curve a controller is sensitive to. That needs a metric this literature does not use: whether two twins pick different actions is not the same as whether picking the wrong one costs anything. We therefore measure *regret*, the utility an rApp loses by planning on one twin and being evaluated on another. Under regret, almost all of the OAI curve’s value collapses onto a single implementation margin δ on the SINR axis. A second constant, a throughput derate κ , converts the twin’s link rate into what a real MAC schedules. The twin needs these two numbers from a real stack and, for this controller, nothing else.

This paper contributes:

- 1) *Two constants, and evidence that they are enough.* We separate a twin’s *reporting* error (what it says) from its *decision* error (what acting on it costs, measured as rApp regret). An uncalibrated Shannon curve inflates throughput by 41% and costs 0.76 nats of proportional-fair utility. A single fitted implementation margin $\delta = 4.71$ dB reduces these to 1.5% and 0.04 nats; a fitted attenuation factor does not (-6.0% , 0.80 nats). The curve’s non-uniform shape buys nothing further for this controller. We also show the customary “fraction of layouts whose

decisions differ” (33–42% for *every* surrogate) cannot distinguish them—only regret can.

- 2) *A constant, emulator-independent MAC derate.* On a live OAI Standalone stack at matched bandwidth (106 PRB, 30 kHz, 38.16 MHz), the twin’s link-level rate overpredicts what OAI’s MAC schedules by κ^{-1} with $\kappa = 0.470 \pm 0.004$ —constant to 0.8% across MCS 0–28, of which the TDD downlink duty 0.743 is exactly computable. Constancy is the point: η cancels in ratios, so the twin’s *relative* per-UE steering-gain prediction matches the real MAC to 1.2 ± 5.7 points, while delivered application goodput sits a further 80% down and grows with SINR—the emulator’s compute ceiling, not the twin’s abstraction error. Conflating the two is what makes “fidelity gap” numbers irreproducible.
- 3) An open, CPU-only, multi-cell/multi-user digital twin (OpenStreetMap $\rightarrow$ Sionna RT $\rightarrow$ per-RE SINR $\rightarrow$ EESM $\rightarrow$ per-UE/cell throughput, swappable network source) and three twin-to-real-stack bridges: ray-traced channel injection into a live gNB–UE link, a per-UE interference-as-noise-floor mapping, and *per-cell* injection on a two-DU stack—exploiting the UE’s role as rfsimulator server, so the k -th DU’s waveform is filtered by `rfsimu_channel_uek` and the DUs’ signals are physically summed: combined-waveform inter-cell interference, not a noise proxy.
- 4) *A null control, and a retraction.* Twin-versus-real comparisons are easy to fabricate accidentally. Give two real cells the *same* channel and the true effect is exactly zero; the natural goodput probe then reports $-21.0 \pm 1.4\%$. We use this to retract a multi-cell gap we previously reported, and state plainly that a controlled multi-cell measurement does not close on this emulator.
- 5) A 24-run, 3-city evaluation of the rApp reporting variance and outage, plus a load-sensitivity sweep. rApp gain tracks peak cell load ($r=0.45$) more than baseline outage ($r=0.27$)—the reverse of what the same experiment reported when the twin sampled a 3.84 MHz sub-band, a reminder that L2S details propagate into conclusions about the controller.

II. RELATED WORK

Open RAN digital-twin platforms. Table I positions the closest open efforts on four axes. OWDT [2] pairs OAI with Sionna RT for a faithful single gNB–UE link, but is single-cell and monitor-only. Tiny-Twin [3] runs a real OAI stack for multiple UEs with a RIC loop, single-cell from replayed traces. Orange’s RIC-TaaP/ns-O-RAN and OpenTwin [4] close rApp loops across many cells—OpenTwin adds twin-to-real KPM self-correction—but on ns-3’s abstracted PHY. OAI+FlexRIC work [5] closes handover loops across real cells over the air, without a ray-traced channel. Commercial twins close the loop but are closed-source.

Link-to-system abstraction. System-level simulation [6] relies on L2S mapping: a link-level simulator characterises SINR-to-BLER per MCS, and an effective-SINR mapping

TABLE I
OPEN RAN DIGITAL-TWIN PLATFORMS ON FOUR AXES.

Platform	Real stack	Site RT	Multi cell	Closed loop
OWDT [2]	✓	✓	×	×
Tiny-Twin [3]	✓	×	×	✓
Orange ns-O-RAN / RIC-TaaP	×	✓	✓	✓
OpenTwin [4]	×	×	✓	✓
OAI+FlexRIC [5]	✓	×	✓	✓
OAI VRTSim / RT emulator	✓	✓	×	×
This work	✓	✓	✓	✓*

^areplayed traces; ^bns-3 abstract PHY; ^cover-the-air; ^dmulti-node on roadmap; *SA loop closed & measured on live stack (scripted F1 HO for multi-cell).

(EESM [7], [8]) collapses a frequency-selective SINR vector into an AWGN-equivalent value. Our lookup is measured from OAI’s own PHY rather than from idealised curves. Our interference-as-noise-floor bridge is a site-specific generalisation of the 3GPP OFDMA channel noise generator (OCNG) [9]: we derive each UE’s interference from the ray-traced neighbour channels.

O-RAN control loops. The O-RAN architecture [10], [11] externalises RAN control: xApps on the near-RT RIC (via E2) and rApps on the non-RT RIC (via O1/A1). Traffic steering via cell-individual offsets (CIO) is a canonical rApp use case [1], which a digital twin lets one validate before deployment.

III. DIGITAL TWIN ARCHITECTURE

The platform is a pipeline with two consumers of one physics substrate (Fig. 1): a fast analytical layer covering every cell and UE, and a real-stack layer grounding sampled links in OAI. Both are driven by the same ray-traced channel; the network data source is swappable, so synthetic and real deployments share one path.

Geometry. A WGS84 bounding box drives an OpenStreetMap (Overpass) query; footprints are extruded into a 3D Mitsuba scene with ITU materials.

Network. Cells and UEs come from a swappable source: a seeded random generator (N sites $\times$ 3 sectors, uniform UE drops), or a CSV reader for real cell-site coordinates (e.g. OpenCellID). Both feed the same pipeline, so the twin can be calibrated on synthetic data and run on real sites without code changes.

Channel. Sionna RT [12] places one transmitter per cell (3GPP sector pattern, planar array, downtilt) and one receiver per UE, and computes the channel frequency response of every cell-to-UE link over an OFDM resource grid, capturing reflections, diffractions and scattering from the building geometry. Interaction depth is 3.

IV. OAI-GROUNDED KPI ENGINE

Let $\mathbf{H}_{u,c}(k) \in \mathbb{C}^{N_r \times N_t}$ be the ray-traced channel of link $c \rightarrow u$ on resource element (RE) k , P_c^{tx} the cell transmit power, and $P_{u,c}^{\text{rx}} = P_c^{\text{tx}} \|\mathbf{H}_{u,c}\|_F^2$ the wideband received power (bar = average over antennas and REs). Each UE associates to the cell

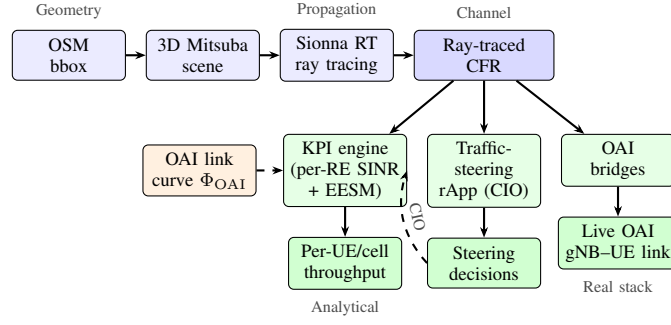


Fig. 1. Architecture. OpenStreetMap geometry and a swappable network source feed Sionna RT; the ray-traced channel drives (left) an analytical multi-cell KPI engine and the traffic-steering rApp, and (right) a live OAI link over the exact channel with a per-UE interference noise floor. The OAI-measured link curve Φ_{OAI} grounds the analytical throughput in a real receiver.

maximising received power biased by a cell-individual offset (CIO), the O-RAN control knob:

$$s(u) = \arg \max_c P_{u,c}^{\text{rx}} \cdot 10^{\text{CIO}_c/10}. \quad (1)$$

Each cell serves one UE at a time under airtime sharing, so the optimal single-stream precoder is the matched filter and the serving per-RE gain is the largest singular value $\sigma_{\max}$. With flat per-RE transmit power p_c , neighbour load $\rho \in [0, 1]$ and thermal noise $N_0 = kTB \cdot \text{NF}$:

$$\gamma_u(k) = \frac{p_s \sigma_{\max}(\mathbf{H}_{u,s}(k))^2}{\rho \sum_{c \neq s} \frac{p_c}{N_t} \|\mathbf{H}_{u,c}(k)\|_F^2 + N_0}. \quad (2)$$

The per-RE SINRs are compressed to one effective SINR by exponential effective SINR mapping (EESM), preserving the ray tracer's frequency selectivity:

$$\gamma_u^{\text{eff}} = -\beta \ln\left(\frac{1}{K} \sum_k e^{-\gamma_u(k)/\beta}\right), \quad (3)$$

with β set per modulation order [8] and one scale parameter for calibration against OAI ground truth.

OAI-measured link curve. The effective SINR is mapped to an MCS and spectral efficiency, $(m_u, \eta_u) = \Phi_{\text{OAI}}(\gamma_u^{\text{eff}})$, through a curve *measured offline from OAI's nr_dlsim* (real LDPC decoding and rate matching), so it carries a real receiver's implementation losses: decoding margins, rate-matching overhead, and non-uniform SINR spacing between MCS transitions. It is measured once by sweeping $\text{SNR} \times \text{MCS}$ and recording the lowest SNR at which first-transmission BLER stays within the 10% target.

At 10 dB SINR the OAI curve delivers 2.16 bits/s/Hz—38% below Shannon's 3.46—and the MCS transition spacing is non-uniform: the QPSK→16QAM boundary spans 2.2 dB against ~1 dB for adjacent QPSK steps. Whether that *shape* matters to a controller, as opposed to the overall 38% level, is the question Sec. VIII-B answers.

On a static full-buffer snapshot, proportional-fair scheduling reduces to equal airtime, so the per-UE throughput is

$$R_u = \frac{B_s}{K_s} \eta_u (1 - \text{BLER}_{\text{tgt}}), \quad (4)$$

where B_s is the serving cell bandwidth and K_s its number of attached UEs.

V. TRAFFIC-STEERING RAPP

The rApp adjusts the CIO vector to maximise the proportional-fair utility $U(\text{CIO}) = \sum_u \log R_u$. Because U is piecewise-constant in the CIOs (it changes only when a UE's serving cell flips), a fixed small step stalls on plateaus. We therefore do coordinate ascent: starting from $\text{CIO} = 0$, sweep the cells, line-searching each cell's CIO over a grid with the others fixed and keeping the maximiser of U , until no cell's line search improves U . This is monotonically non-decreasing in U by construction. Raising a cell's CIO pulls boundary UEs in (absorbing load); lowering it sheds them to neighbours.

Every KPI evaluation reuses the same engine with a `cio_db` vector, so the control loop rides on the exact twin used for the baseline. Its output is a set of per-UE handover decisions a real stack would execute via OI/telnet; it converges in 2–3 sweeps, suitable for the non-RT RIC timescale.

VI. TWIN-TO-REAL-STACK BRIDGES

Three bridges connect the analytical twin to a live OAI stack, grounding per-link and per-cell realism.

Channel injection. A bridge converts a link's ray-traced channel frequency response to a sample-spaced impulse response (IFFT), retains the strongest causal taps, normalises them to unit energy, and maps each tap delay to a sample index. These taps override the rfsimulator's channel through a patch to OAI's channel module (`oai_rt_inject_channel` in `random_channel.c`). When `OAI_RT_TAPS` is set, the gNB-UE link runs over the site-specific channel. Injection is confirmed in the gNB log (`[RT] injected 4 taps ... channel_length=49`); the link then runs its PHY/MAC over the ray-traced channel.

Interference-as-noise-floor. A live single-gNB rfsimulator link has no notion of the other cells a UE hears. A second bridge computes, per UE, the inter-cell interference and expresses it as a noise-floor rise $\Delta N_u = 10 \log_{10}((I_u + N_0)/N_0)$, where I_u is the load-scaled sum of non-serving cells' received power. This is written as a per-client OAI noise power, so a single-link run reflects its multi-cell context—a site-specific generalisation of OCNG [9].

Per-cell channel injection. With two DUs the noise-floor proxy is unnecessary. In the handover topology the UE is the rfsimulator *server*, so the k -th DU to connect is filtered by channel model `rfsimu_channel_uek` and `rxAddInput` accumulates every connection into the UE’s receive buffer. One taps block per DU therefore presents each cell’s own ray-traced link gain, and the non-serving cell’s waveform is summed in as genuine co-channel interference. We also make DU1 co-channel with DU0 (identical but for PCI/cell identity); the stock configuration places them on different carriers.

VII. EXPERIMENTAL SETUP

Scenes. Berlin-Mitte (2,623 buildings), Paris-Opéra (121), Manhattan-Midtown (119); geometry fetched/built once per scene and reused across seeds.

Random layouts. Eight seeded layouts per scene—24 runs (323 s on one CPU server). Radio and topology: 3.5 GHz, $B = 38.16$ MHz (106 PRB at 30 kHz SCS, matching the live OAI cell), 46 dBm per cell, 4×1 BS array and single UE antenna, 7 dB receiver noise figure, ray-tracing interaction depth 3, 9 cells and 30 UEs per layout, default inter-cell load $\rho = 1.0$. The CFR is exported over all 1272 subcarriers, so the per-RE SINR and its EESM compression cover the cell’s full band.

Configuration sweep. Neighbour load factor in $\{0, 0.25, 0.5, 0.75, 1.0\}$ on a cached channel per scene.

Metrics. Per-UE downlink throughput (median, mean); *served cell-edge* rate, the 5th percentile among UEs with non-zero throughput (a raw 5th percentile collapses to zero once outage exceeds 5%); *outage*, the fraction of UEs with zero throughput; Jain’s fairness index; peak cell load. rApp gains are mean $\pm$ std over the eight seeds per scene, pooled across all 24 runs; undefined when a baseline metric is zero.

VIII. RESULTS

A. Traffic-steering rApp

Across 24 runs, relative to strongest-cell association (Table II): the rApp helps the users a strongest-cell policy under-serves (median, cell-edge, fairness) and sheds peak load. In absolute terms the median per-UE rate rises from 16.0 to 18.3 Mbps and the served cell-edge rate from 5.8 to 7.4 Mbps, while the mean barely moves ($22.1 \rightarrow 22.6$ Mbps). Three honest observations: aggregate rate is nearly unchanged (+2.2%; the proportional-fair objective trades sum for fairness), outage is *exactly* unchanged in every one of the 24 runs (steering relocates covered users, it does not create coverage), and variance is high and layout-dependent—motivating multi-seed evaluation.

Per scene (Table II): Paris has the highest outage (25.8%) and the largest median gain among covered users; Berlin, the densest, has the lowest outage (10.0%) and sheds the most peak load (−1.1 UEs); Manhattan gains most in fairness (+16.7%). The direction is robust across all three, the magnitude strongly deployment-dependent—exactly what a single-scenario study would misrepresent.

TABLE II
RAPP GAINS VS. STRONGEST-CELL (MEAN $\pm$ STD OVER SEEDS).

Metric	Overall	Berlin	Paris	Manh.
Median tput gain	$12.0 \pm 12.8\%$	8.7%	14.2%	13.3%
Served edge gain	$28.6 \pm 37.2\%$	35.0%	32.7%	18.2%
Jain fairness gain	$7.1 \pm 13.0\%$	4.6%	0.0%	16.7%
Peak load (UEs)	−0.67	−1.12	−0.38	−0.50
Mean tput gain	$3.0 \pm 7.2\%$	3.8%	4.1%	1.1%
Outage change	0.0 pts	0.0	0.0	0.0
Baseline outage	17.6%	10.0%	25.8%	17.1%

B. What a twin says vs. what acting on it costs

The 38% link-curve gap of Sec. IV is real, but does a controller care? Hold geometry, ray tracing, association and interference fixed and change *only* the link curve. The *reporting* error is how far a surrogate twin’s predicted throughput sits from the OAI-grounded twin’s. The *decision* error is *regret*: take the CIO the rApp selects on the surrogate, evaluate it on the OAI-grounded twin, and compare with the CIO it would have chosen had it planned there,

$$\mathcal{R} = U(\text{CIO}_{\Phi_{\text{OAI}}}^* | \Phi_{\text{OAI}}) - U(\text{CIO}_{\Phi_{\text{sur}}}^* | \Phi_{\text{OAI}}) \geq 0, \quad (5)$$

with U the proportional-fair objective the rApp maximises. Reporting error is what you get wrong by reading a number off the twin; regret is what you pay for acting on it. We compare three surrogates over the same MCS/SE set, differing only in the SINR at which each MCS becomes available: the Shannon-optimal staircase; a fitted SINR offset $\Phi(\gamma) = \log_2(1 + \gamma/\delta)$, $\delta = 4.71$ dB; and a fitted attenuation $\Phi(\gamma) = a \log_2(1 + \gamma)$, $a = 0.606$. The last two are what a practitioner calibrates to link-level data with one parameter, and are the honest baselines—the Shannon staircase is a straw man no simulator uses.

The two errors turn out to be close to independent (Fig. 2). The idealized curve inflates throughput by 41.3% and costs 0.76 nats of utility. One fitted number, δ , removes almost all of both: reporting error falls to 1.5% and regret to 0.04 nats—a $19\times$ reduction, and small enough that planning on it is, for this rApp, equivalent to planning on the real PHY curve. The attenuation model reports well (−6.0%) yet is as costly to act on as no calibration at all (0.80 nats): reporting accuracy does not imply decision quality. Most tellingly, the statistic this literature usually reports—the fraction of layouts on which the two twins select different steering actions—is 37.5%, 33.3% and 41.7% respectively. It cannot tell the good surrogate from the bad ones. Only regret can. What the twin needs from a real PHY is therefore δ ; the curve’s non-uniform MCS spacing, which is what measuring OAI actually buys beyond one scalar, does not change what the controller does.

C. The real fidelity gap

The analysis above compares two *models*; we now close the loop on the real stack and measure the gap directly. We stand up a full OAI 5G Standalone system on the same CPU host—core (AMF/SMF/UPF), gNB, and a UE that registers,

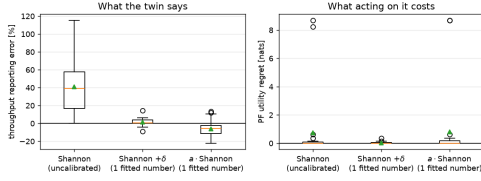


Fig. 2. Reporting error (left) and the cost of acting on it (right), per surrogate twin, over 24 runs. A single fitted implementation margin δ collapses both; a fitted attenuation collapses only the first.

establishes a PDU session and carries user-plane traffic. On a single gNB we ground the *link* dimension: we pin the downlink operating point by capping the scheduler MCS over a clean channel, sweep the cap across MCS 0–28, and measure end-to-end goodput at each fixed MCS, which indexes the same OAI curve the twin uses (a shared SINR axis). Two details decide the answer. The live cell is 106 PRB at 30 kHz SCS, i.e. $B = 38.16$ MHz occupied, not the twin scenario’s 20 MHz, so the twin’s rate must be evaluated at the cell’s own bandwidth. And a saturating UDP probe is unusable: in reverse mode *iperf3* reports the *sending* rate, so with the scheduler pinned to MCS 4 (a 9.8 Mbps MAC) an 8 Mbps offer reads exactly 8.00 Mbps at $\sim 0\%$ loss. TCP cannot exceed the channel but needs tens of seconds to leave slow start, so we settle and measure its plateau.

Two references matter, and conflating them is what makes “fidelity gap” numbers irreproducible. Against **what OAI’s MAC schedules**—real 5G-NR scheduler code paying TDD duty, DMRS, PDCCH and HARQ—the twin’s link rate $B\eta_{\text{OAI}}(1 - \text{BLER})$ is high by a *constant* factor: $\kappa = 0.470 \pm 0.004$ (a 52.9% inflation, ranging only 52.4–53.6% over MCS 0–28; CV 0.8%). Of that, the TDD downlink duty $104/140 = 0.743$ is exactly computable; the residual 0.633 is physical-channel and scheduler overhead. Against **delivered application goodput** the twin is high by a median 80%, and this gap is *not* constant: it grows from 72% at MCS 0 to 87% at MCS 28 (derate CV 27%). At MCS 28 the twin predicts 190.8 Mbps, TDD duty allows 141.7, OAI’s MAC schedules 88.9, the application receives 24.9. Only the first two steps are modelling; the last is the rfsimulator’s real-time compute ceiling—an artefact of a CPU software radio, not a property of 5G (Fig. 3, left).

Constancy is not a detail: it is the second constant. A multiplicative κ cancels in any ratio, so it should not perturb the twin’s *relative* predictions at all. We test that prediction by propagating each curve through the rApp over eight layouts (240 per-UE evaluations, 19 steered UEs), indexing every curve by the twin’s own operating MCS—the twin’s rate is a staircase in SINR, so interpolating a measured curve along SINR instead would compare a staircase with a ramp and manufacture a gap between the knots. We keep the twin’s airtime share B_c/K_c in *both* rates, since that term is most of the rApp’s benefit and cancels in the difference. Against the MAC the twin’s per-UE steering gain is right: 126.6% predicted vs 125.4% scheduled, a gap of $+1.2 \pm 5.7$ points

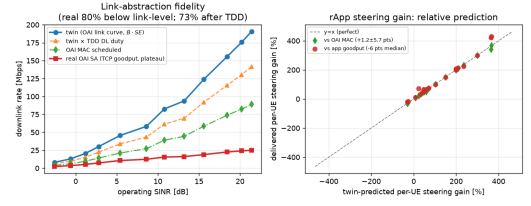


Fig. 3. Single gNB, matched bandwidth. Left: at matched operating SINR—the twin’s link-level rate, that rate scaled by the TDD DL duty, OAI’s scheduled MAC throughput, and delivered goodput. The twin-to-MAC ratio is constant ($\kappa = 0.470 \pm 0.004$); the MAC-to-application spread is the emulator’s ceiling and grows with SINR. Right: per-UE rApp steering gain. Against the MAC the twin sits on $y=x$ ($+1.2 \pm 5.7$ pts); against delivered goodput it does not.

(Fig. 3, right). Against delivered goodput it is not, by a median -6 points (mean -12.4 , std 20.7; we do not headline a mean whose spread exceeds it)—and that residual is the SINR-dependent emulator ceiling, not the abstraction. An OAI-grounded twin, corrected by one scalar κ , therefore predicts what a real 5G MAC will schedule, in absolute terms and in the ratios a controller consumes.

D. Multi-cell closed loop, and what it does not yet measure

Two real cells (CU plus two F1 DUs, PCI 0/1) match a two-sector twin (20 UEs). The rApp’s association decision executes as a real inter-cell handover via scripted CU telnet (`ci trigger_f1_ho`), PDU session preserved, flow continuing on the target cell: the loop is closed. *Measuring* the delivered gain is another matter. Our first attempt reported the twin’s per-UE gain (66%) overshooting the delivered gain (9.0%) by 57 ± 165 points, and blamed the DUs’ clean channel. That does not survive a null control: give the cells the *same* channel and the true effect is exactly zero, so whatever the harness reports is bias. It reports $-21.0 \pm 1.4\%$ (Fig. 4). The cause is timing—a 6 s TCP flow started 3 s after each handover, inside both slow start and the link-adaptation ramp—and since `trigger_f1_ho` always moves DU0→DU1, measurement order and cell identity were perfectly confounded, so the reported sign followed the direction the rApp happened to choose (on identical cells the probe yields mean DL MCS 16.5 on PCI 0 vs 10.7 on PCI 1). Probing at TCP’s plateau gives $-2.4 \pm 2.9\%$, consistent with zero, MCS 28 on both. We withdraw the 57 ± 165 number.

With the instrument fixed we ground the channel (Sec. VI). Injection works: every layout’s UE log confirms both per-cell models carrying exactly the twin’s ray-traced gains, and the UE reaches MCS 27 on the twin’s stronger cell. Two obstacles remain, neither the twin’s. First, with channel models on both rfsim connections the loop destabilises: over three layouts two F1 handovers never complete and the third only through 70 UE re-synchronisations, so its goodput measures the emulator’s recovery, not the channel. Second, rfsim’s noise floor is a config knob, not $kTB \cdot \text{NF}$: injection sets the *relative* per-cell gain, not the absolute SINR. Sweeping it from -22 to -2 dB with one cell muted, the operating MCS goes

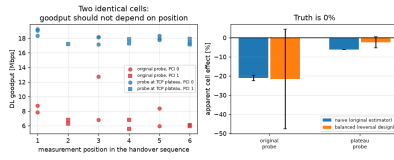


Fig. 4. Null control: both cells carry the *same* clean channel, so the true cell effect is 0%. Left: with the original probe, goodput depends on position in the handover sequence, and PCI 0—measured at odd positions—separates from PCI 1. Right: the original estimator reports $-21.0 \pm 1.4\%$; the plateau probe gives $-2.4 \pm 2.9\%$.

8, 19, 14, 18, 14, 0—non-monotone, hence not invertible to a target operating point. And at the idle-neighbour condition a one-UE stack presents, the twin puts both cells at MCS 28 and predicts *no* link-level difference in two of three layouts. A controlled multi-cell gap needs absolute operating-point matching, which this emulator cannot deliver; we report the negative result rather than a number.

E. Coverage, outage, and load sensitivity

Mean UE outage is 17.6% at full neighbour load (Berlin 10.0%, Manhattan 17.1%, Paris 25.8%). rApp median gain rises with peak cell load ($r=0.45$, $R^2=0.20$) and only weakly with baseline outage ($r=0.27$): load imbalance, not coverage, is what steering has to work with. Lowering neighbour load from 1.0 to 0 raises mean throughput $1.68\text{--}1.99\times$, separating a coverage-limited tail that steering cannot fix from an interference-limited body where load balancing acts. An earlier version of this experiment, whose CFR sampled only a 3.84 MHz sub-band and applied the resulting EESM bandwidth, reported the opposite ordering ($r=-0.42$ on outage, $R^2=0.04$ on load). Exporting the cell’s full 38.16 MHz reversed it: the L2S pipeline does not merely rescale a twin’s numbers, it changes which deployment property the twin says a controller is sensitive to.

IX. DISCUSSION AND LIMITATIONS

We state these plainly. (1) The OAI Standalone loop is closed on one gNB (link calibration, Sec. VIII-C) and on two cells with scripted F1 handovers (Sec. VIII-D); autonomous E2-RC steering remains future work. (2) The single-gNB numbers are measured against *this* emulator, which delivers only 24.9 of the 88.9 Mbps its own MAC schedules at MCS 28; the twin’s error is bounded by the MAC line, not the goodput line—why we headline κ , not the 80%. (3) Per-cell injection is verified and rfsim sums the DUs’ waveforms—inter-cell interference is physically combined, superseding the noise-floor rise ΔN_u the analytical engine uses—but the multi-cell measurement does not close (Sec. VIII-D), and the injected taps carry each link’s gain, not its delay profile. (4) The two constants are established for *one* controller (CIO traffic steering) on a proportional-fair objective. Whether one scalar δ suffices for controllers that read MCS directly—link adaptation, MCS-aware schedulers—is open; our claim is a measured result for this rApp, not a general law. (5) The evaluation is a static full-buffer snapshot, single carrier, downlink,

and validation is stack-versus-model; mobility, uplink, field calibration and real cell-site databases are the next steps.

X. CONCLUSION

We asked how much of a real physical layer a 5G digital twin actually needs, and measured the answer on OpenAir-Interface. For a proportional-fair traffic-steering rApp, two constants suffice. An implementation margin $\delta = 4.71$ dB on the SINR axis cuts an uncalibrated twin’s throughput reporting error from 41.3% to 1.5% and its control regret from 0.76 to 0.04 nats; the measured curve’s non-uniform shape buys nothing further. A throughput derate $\kappa = 0.470 \pm 0.004$ —constant to 0.8% across MCS 0–28—converts the twin’s link rate into what a real 5G MAC schedules, and being constant it cancels in ratios, so the twin’s *relative* steering-gain predictions are already right to 1.2 ± 5.7 points. Below the MAC lies an 80% shortfall that grows with SINR: the software radio’s compute ceiling, not the twin’s error. Two methodological points earn their place beside the constants. The customary “decisions differ” statistic (33–42% for every surrogate we tried) cannot separate a faithful twin from an unfaithful one; only regret can. And a twin-versus-real comparison must be run against a null control—on two *identical* real cells the natural goodput probe reports $-21.0 \pm 1.4\%$, which is how we came to retract a multi-cell gap we had previously published. Code, data and OAI logs reproduce every number (Apache-2.0).

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